

SAHIL AHAMED SHAIK



UNSW Robotics Master's Student · AI & Autonomous Systems

5+

Major Projects

2

Internships

3

Conferences

1

AIP Publication



Master's in Robotics — UNSW Sydney | B.Tech Mechatronics — Hindustan Institute of Technology & Sciences (HITS)

ROS 1 & 2

SolidWorks

Python

C++

TensorFlow

ANSYS

SLAM

Unity AR/VR

MATLAB



Professional Experience

Robotics Intern

Abhyaz

- Built hands-on expertise in ROS architecture, topics & launch files
- Mastered SolidWorks & Fusion 360 for parametric 3D modeling
- Self-directed learning approach to tackle advanced robotics challenges



ROS Intern

Edgeforce Solutions

- UGV path-planning: Move Base (DWA) & custom A* with SLAM
- 3D point cloud obstacle detection via LiDAR + camera integration
- CBM system design, exoskeleton work & airport tug CAD (PLANSYS2)

Key Deliverables from Edgeforce



Gazebo & RViz
Simulations



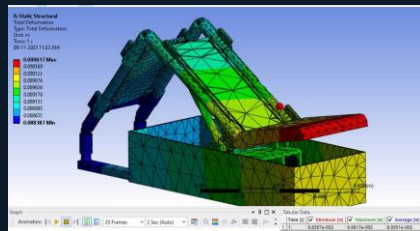
UGV Physical
Deployment



Airport Tug
CAD Design



PLANSYS2
ROS 2 Task Planner



FEATURED PROJECT · 01

Knee Rehabilitation Robot

AI-Augmented Physiotherapy System

SolidWorks

ANSYS FEA

TensorFlow / CNN

Patent Pending

PROJECT OVERVIEW

Autonomous physiotherapy robot delivering targeted knee rehabilitation, replacing clinician presence with precise 2-DOF arc motion and AI-powered diagnosis.

5+

SolidWorks Designs

3

Physical Prototypes Built

56%

CNN Model Accuracy

1★

AIP Publication 2023

TECHNICAL HIGHLIGHTS

- ANSYS static structural FEA: max deformation 0.09 m within safe patient limits
- CNN (ReLU, 100 epochs) trained across 5 knee osteosis severity classes
- Presented at RIACT 2022, RANE 2023 & AIIMS-3.0 — patent filing in progress



UGV Autonomous Path Planning

Method A: ROS Move Base (DWA)

Used the ROS navigation stack with Dynamic Window Approach local planner and trajectory planning for obstacle-reactive path following. Integrated global & local cost maps.

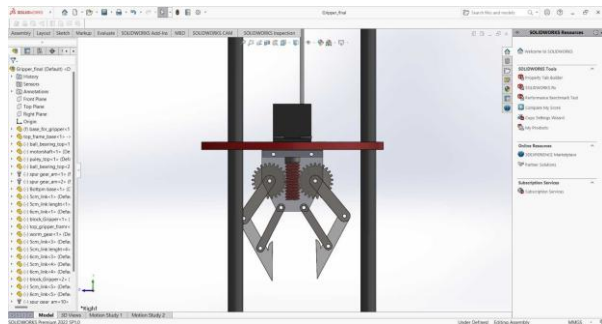
- 3D Point Cloud (LiDAR) for obstacle detection
- Camera for region-of-interest targeting
- Validated in Gazebo simulation first
- Deployed on physical UGV platform

Method B: Custom A* + SLAM

Built a bespoke A* pathfinding algorithm with SLAM-based occupancy mapping, offering full control over global and local path generation with waypoint-based static routing.

- RViz for 2D/3D map visualization
- Occupancy grid with static waypoints
- SLAM for real-time environment mapping
- Airport tug extension via PLANSYS2

Rescue Robotics & Swarm Intelligence



Borewell Rescue Robot

Child rescue from borewell incidents via soft-gripper robotic system.

- Soft grippers to safely extract child without injury
- 2 CAD designs created; 1 physical prototype developed
- Focused on simplicity, affordability & rapid deployment
- Market-available components used for reproducibility
- Full SolidWorks modelling with dimensional accuracy



Swarm Robot

Multi-agent coordination in a virtual environment using Webots simulator

- Studied swarm algorithms & inter-robot communication
- Webots simulator: Python-based controller programming
- Central world node used as communication medium
- Cumulative X-Y plane movement for coordinated motion
- Identified need for interrupt-based obstacle avoidance

Sanitization Robot & Other Work



Autonomous Sanitizing Robot

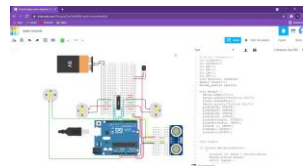
Prototype for automated shop sanitization during the COVID-19 pandemic.

Ultrasonic sensors for obstacle avoidance.
IR-triggered spray mechanism near shelves.
Tinkercad-based circuit and code validation.

Load-tested chassis for onboard water tank.

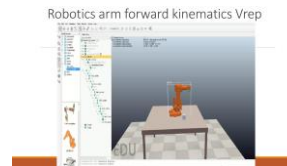
Additional Prototyping & Simulation Work

Selected screenshots from simulation, planning and perception work.



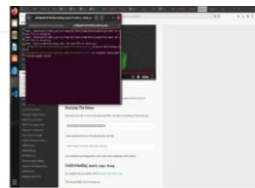
Circuit Simulation

Tinkercad wiring and code validation.



V-REP Kinematics

Forward kinematics of a 7-DOF robotic arm.



MoveIt Planning

ROS MoveIt simulation for manipulator motion planning.



RL & OCR Experiments

Reinforcement learning, OCR and vision prototypes.

Publications & Presentations



Journal Publication

Design and Evaluation of Various Knee Assistive Mechanisms for Rehabilitation

AIP Publishing · November 10, 2023



RIACT 2022

Robotics, Instrumentation, Automation &
Control Technology

RANE 2023

Recent Advancements in New Era
Engineering

AIIMS-3.0

AI in Medical Systems — Knee Rehab
Robot showcase



Patent application and additional journal submission currently in progress for the knee rehabilitation robot design.



Skills & Tools

Core software, CAD, programming and specialization areas.

Robotics Software

- ROS 1 (Noetic)
- ROS 2 (Humble)
- Gazebo Simulator
- RViz Visualizer
- Movelt Motion Planning
- PLANSYS2 Task Planner

CAD & Simulation

- SolidWorks (Advanced)
- Fusion 360
- ANSYS (FEA / Structural)
- MATLAB & Simulink
- V-REP / CoppeliaSim
- Unity (AR / VR)

Programming

- Python (Primary)
- C++
- TensorFlow 2 / Keras
- OpenCV
- NumPy & Matplotlib
- Embedded C (Arduino)

Specializations

- SLAM & Path Planning
- Computer Vision / OCR
- Reinforcement Learning
- Exoskeleton Design
- Swarm Robotics
- Pneumatics (SMC)



Certifications & Training

Courses and training supporting robotics, AI and simulation.



Completed

ROS for Beginners

Basics, Motion & OpenCV

Udemy



Completed

TensorFlow 2.0

Deep Learning & AI

Udemy



Completed

Intro to AI & Python

Machine Learning Basics

Udemy



Completed

MATLAB & Simulink

Onramp Training

MathWorks



Completed

SMC Pneumatics

3-Day Intensive Training

SMC Corporation



Completed

Reinforcement Learning

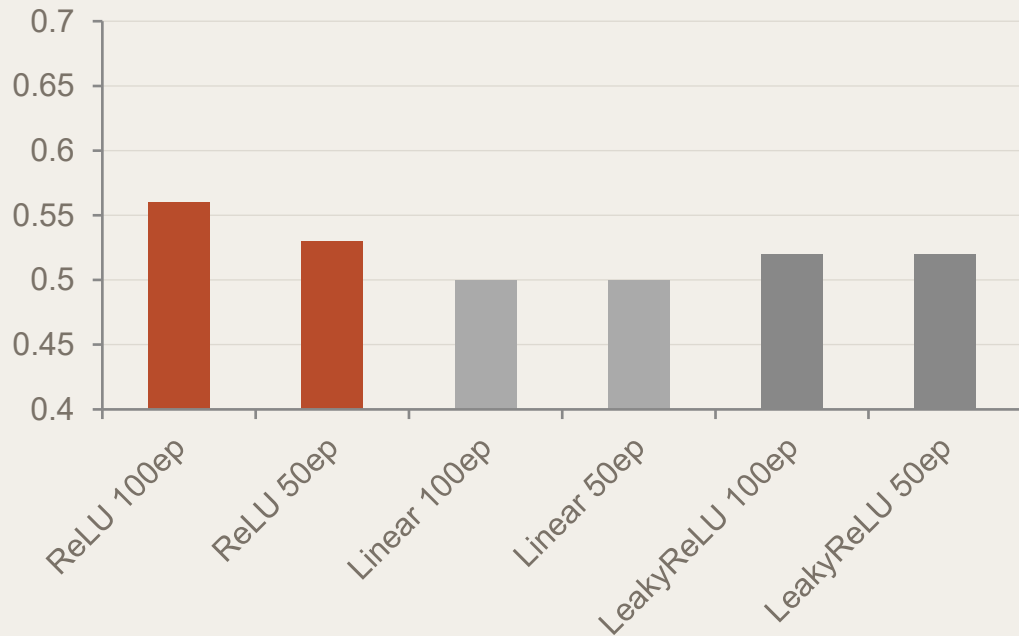
Advanced Algorithms

Udemy



AI Algorithm: Knee Osteosis Detection

Accuracy by Activation Function



MODEL PERFORMANCE SUMMARY

Activation	Epochs	Test Loss	Accuracy
ReLU	100	3.48	56% ★
ReLU	50	2.82	53%
Linear	100	3.71	50%
Linear	50	3.11	50%
LeakyReLU	100	4.58	52%
LeakyReLU	50	3.08	52%

Best model: ReLU with 100 epochs achieved 56% accuracy across 5 severity classes (Normal, Doubtful, Mid, Moderate, Severe). Further gains possible with more training data.

Education & Overview

Current Master's Study

Master's in Robotics

UNSW Sydney · Australia

Focus: robotics, autonomous systems, ROS, navigation and manipulation.

Undergraduate Degree

B.Tech — Mechatronics Engineering

Hindustan Institute of Technology & Sciences (HITS) · Chennai, India

Foundation in mechanical, electronics, control systems and programming.

5+

Robotics Projects

2

Internships

3

Conferences Presented

1

Publications

5

Certifications

1

Patent Pending

LET'S CONNECT

Sahil Ahamed Shaik

UNSW Robotics Master's Student · Open to robotics internships & research roles



sahil86469@gmail.com



github.com/sahil86469



linkedin.com/in/sahil-shaik-8bb309219

Thank you for your time.